

PROCESS BOOK

1. PROJECT OVERVIEW

This process book documents our journey analyzing earthquake impacts across countries with varying economic conditions, from initial research to final visualization.

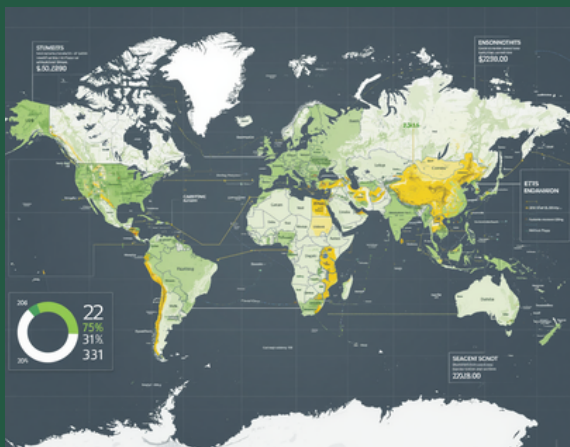
2. RESEARCH PATH

Our research methodology included:

Data Collection: Gathered seismic data and economic indicators from multiple sources.

Comparative Analysis: Studied impact patterns between developed and developing nations.

Visualization Planning: Created initial sketches for data representation.



3. DESIGN CHALLENGES

Throughout the project, we encountered several design challenges:

Data Complexity: Balancing comprehensive data with clear visual presentation.

Scale Differences: Representing vastly different economic scales meaningfully.

Narrative Flow: Connecting earthquake data to economic impact stories.

4. SKETCH EVOLUTION

Milestone 1 Sketches: Initial wireframes and data mapping concepts.

Iterative Refinements: How feedback shaped our visual approach.

Final Design Choices: Color schemes, chart types, and layout decisions.

5. KEY DECISIONS

This project revealed how economic factors dramatically influence earthquake recovery and resilience across nations.

5. TEAM CONTRIBUTIONS

Each team member contributed specific components. Detailed peer assessment and individual contributions are documented in the following pages.

INTRODUCTION

We started with a striking observation:

in 2010, Haiti's magnitude 7.0 earthquake killed over 316,000 people, while Chile's magnitude 8.8 earthquake – releasing 500× more energy just seven weeks later – killed fewer than 600.

The difference was not geology; it was wealth, infrastructure, and preparedness. This contrast became the anchor for our entire project: an interactive scrollytelling visualization exploring why earthquake mortality is shaped by socioeconomic conditions, not seismic force alone.

Our process unfolded in roughly five phases:

1. Data collection & merging.

We gathered the NOAA Significant Earthquake Database (1960–2026, ~2,700 events) and combined it with five socioeconomic indicators from the World Bank and UNDP. Harmonizing country names across six sources (e.g. “Iran” vs. “Iran, Islamic Rep.”) was a significant effort.

2. Exploratory data analysis.

Using a Jupyter notebook we computed correlations, identified outliers, and confirmed our central hypothesis: GDP, HDI, and hospital beds correlate negatively with deaths per earthquake.

3. Sketching & wireframing.

We drafted initial layouts and chart types for each narrative section, iterating on paper and in Figma before writing code.

4. Implementation.

We built the site with Vite, D3.js, Scrollama, and TopoJSON, developing each visualization as an isolated module before integrating.

5. Polish & feedback.

We collected feedback, fixed legend placement, de-collided labels, added responsive design, and refined the storytelling flow.

DATA & EDA

SOURCES

- **Earthquake data** – NOAA NCEI Significant Earthquake Database (2,679 events, 1960–2026).
- **GDP per capita** – World Bank Open Data (NY.GDP.PCAP.CD).
- **Population density** – World Bank (EN.POP.DNST).
- **Urban population %** – World Bank (SP.URB.TOTL.IN.ZS).
- **Hospital beds per 1,000** – World Bank (SH.MED.BEDS.ZS).
- **Human Development Index** – UNDP Human Development Reports 2025.

PRE-PROCESSING

The earthquake TSV contained 39 columns; we extracted country names from location strings (e.g. “PERU: AREQUIPA” → Peru) and filtered to events from 1960 onward with valid magnitudes. Each event was enriched with country-year socioeconomic indicators via left joins. Hospital beds data had the highest missingness (~41%), particularly for low-income countries, which we handled by using nearest-available-year imputation. Country-level aggregates (mean deaths per event, mean magnitude, etc.) were computed for the profile and comparison views. Final processed data was exported as three JSON files: ***earthquakes.json***, ***countries.json***, and ***country_year.json***.

KEY EDA FINDINGS

- Countries with the most earthquakes (Japan, US) are not the deadliest; Haiti and Pakistan show the opposite pattern.
- Deaths are extremely skewed – a handful of mega-disasters account for the majority of recorded fatalities.
- GDP per capita ($r = -0.47$), HDI ($r = -0.37$), and hospital beds ($r = -0.35$) are the strongest negative correlates of log deaths per event; magnitude and depth show weaker associations.

SKETCH & DESIGN EVOLUTION

OVERALL ARCHITECTURE

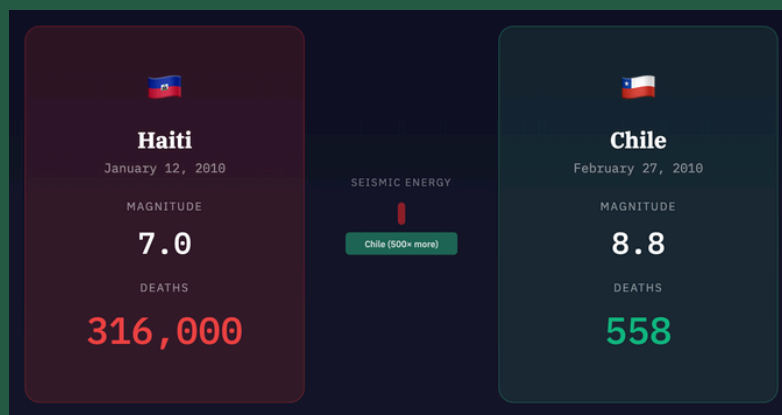
Our initial plan was a standard **single-page scrolling site**. After early prototyping, we restructured into **seven distinct narrative sections**, each combining a sticky graphic with **scrollytelling text steps** powered by Scrollama. This “journalist model” takes the reader from a concrete human story (Haiti vs. Chile) through global patterns, multi-dimensional analysis, and finally free exploration.

SECTION BY SECTION

SECTION 1 - THE HOOK

We needed the opening to be emotional, not statistical.

The side-by-side Haiti/Chile cards with animated death-count counters and a seismic energy bar were designed to make the inequality visceral before any chart appears.

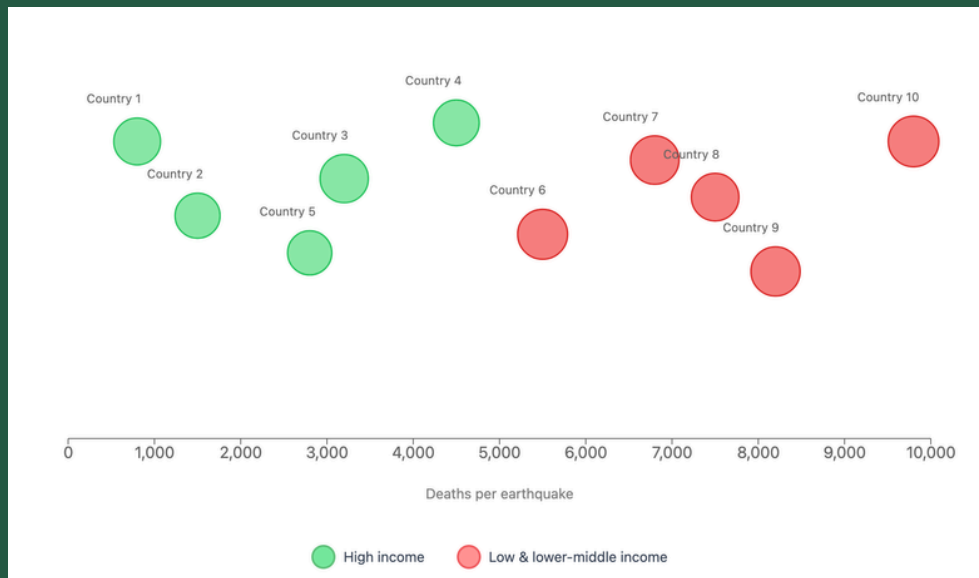


SECTION 2 - THE GLOBE

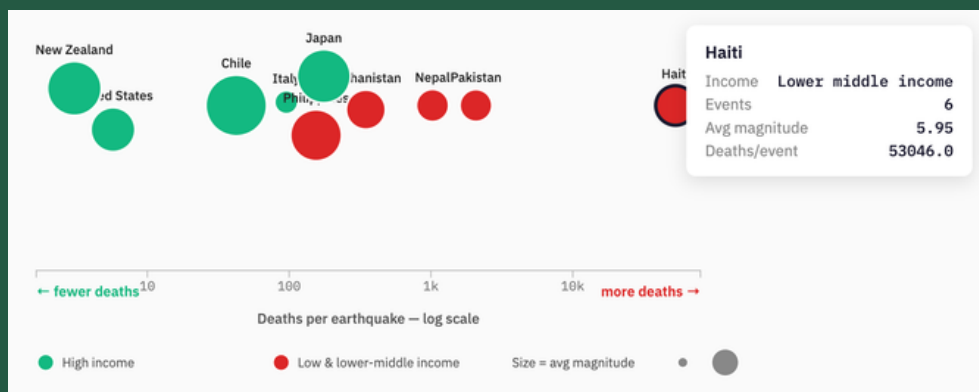
We explored several map projections before settling on an **orthographic globe rendered with D3's geo projection**. The globe rotates to predefined viewpoints on each scroll step, transitioning from showing all earthquakes, to sizing by magnitude, to coloring by death toll, to highlighting the Ring of Fire. An early version plotted static circles; the final version uses animated transitions between encodings.

SECTION 3 - THE INEQUALITY GAP - DOT STRIP

This section went through major iteration. We initially considered a bar chart of deaths by country, but it failed to show how magnitude is orthogonal to mortality. We settled on a single-axis dot strip where each circle is a country, positioned by average deaths per event, sized by average magnitude, and colored by World Bank income group. This encoding makes one point immediately: **wealth sorts; geology does not.**

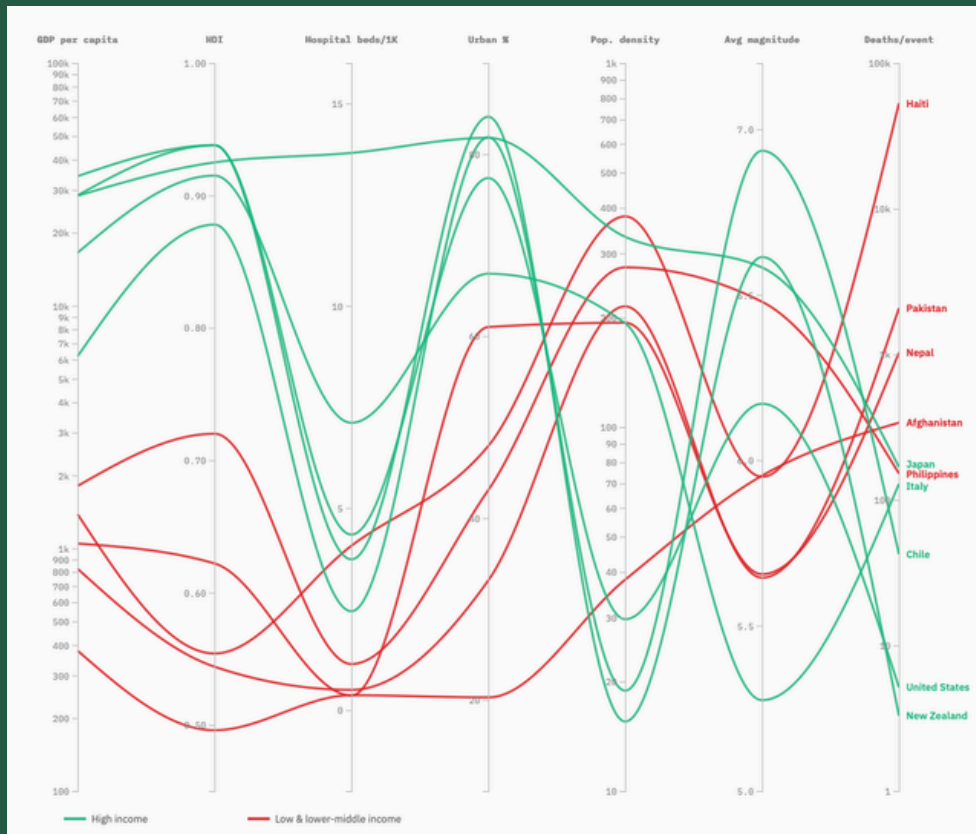


figma graph



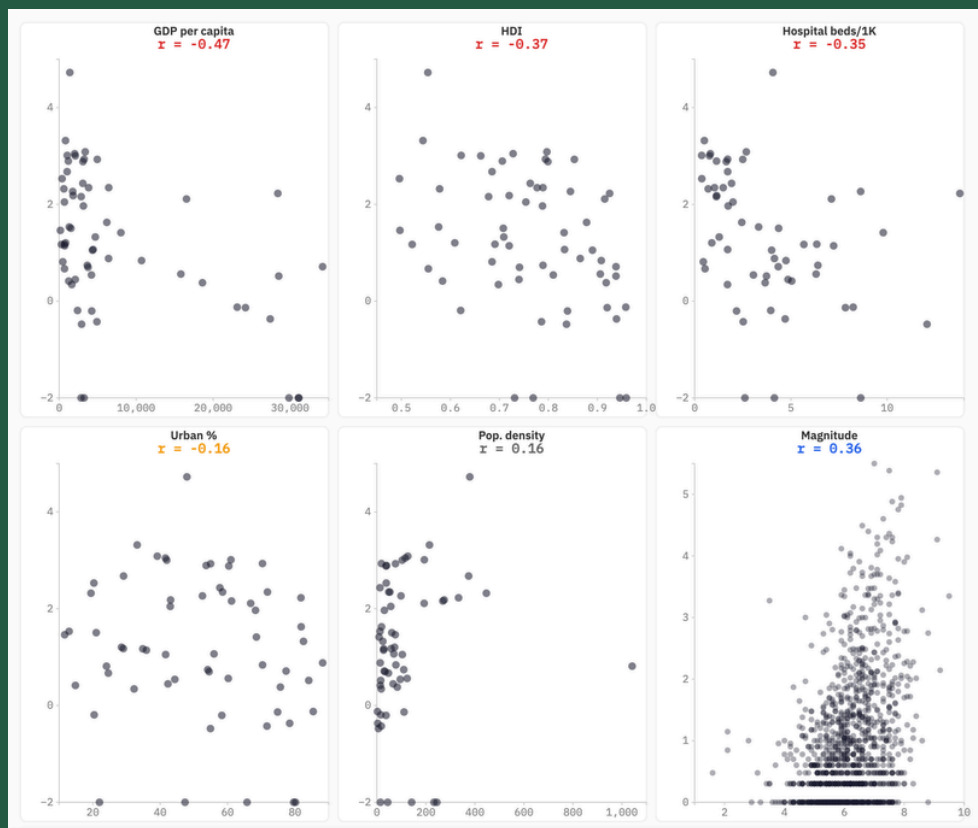
SECTION 4 - COUNTRY PROFILES (PARALLEL COORDINATES)

The parallel coordinates chart threads each country through seven axes: GDP, HDI, hospital beds, urban %, population density, average magnitude, and deaths per event. We used only two colors - green for high-income and red for low/lower-middle income - to make the wealth-mortality gap immediately visible. An early version used a rainbow palette per country, which was visually cluttered and obscured the pattern.



SECTION 5 - CORRELATIONS (SMALL MULTIPLES)

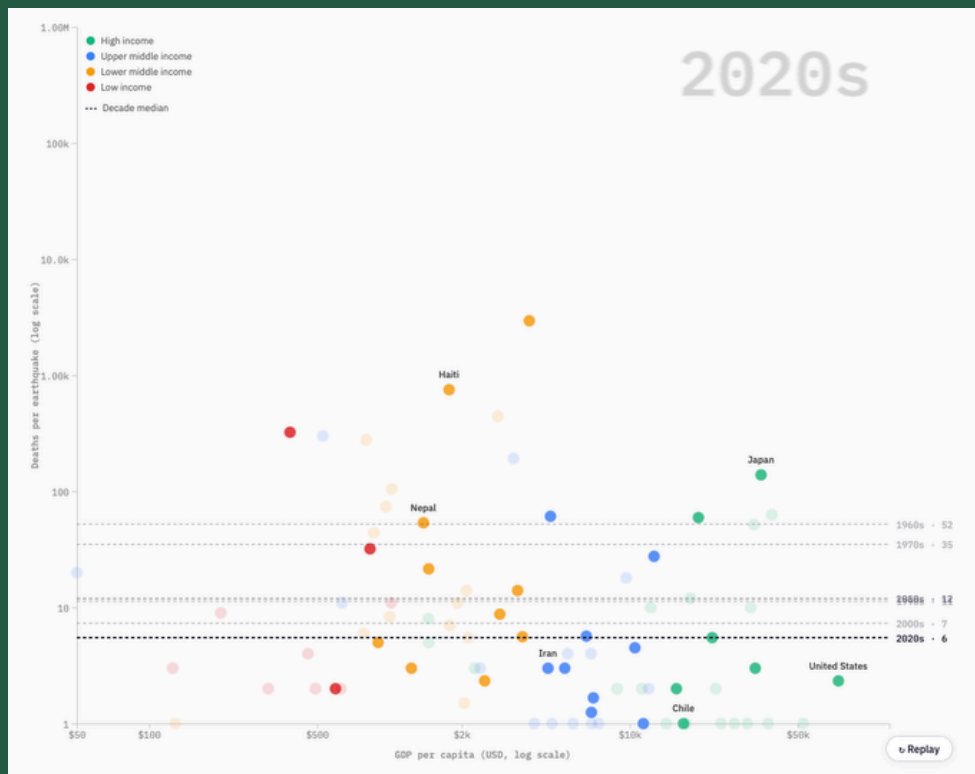
Each small chart plots one socioeconomic indicator against deaths per event, displaying the Pearson r value. We chose small multiples over a single toggle-able chart so readers could compare all factors simultaneously.



SECTION 6 - DECADES (GAPMINDER-STYLE)

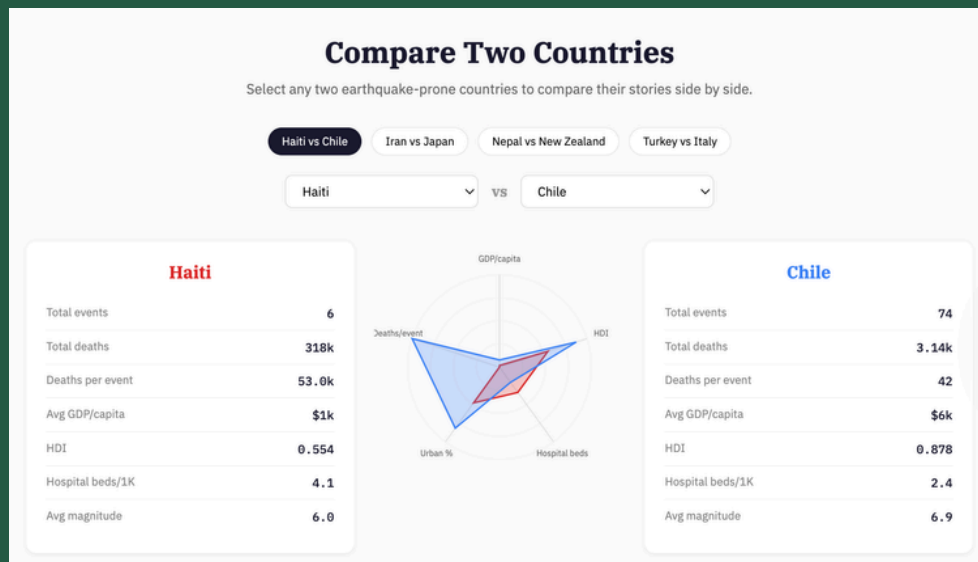
Inspired by Hans Rosling's Gapminder, this animated scatter plot shows **GDP vs. deaths per event** on log-log scales, one decade at a time. Six anchor countries (Haiti, Nepal, Iran, Chile, US, Japan) remain labelled throughout.

Stacked dashed median lines accumulate across decades, showing that the typical fatal earthquake kills far fewer people than it used to. This was the most technically challenging chart due to the animated transitions and median-line layering.



SECTION 7 - COUNTRY COMPARISON

A **free-exploration section** with dropdown selectors and preset pairs (Haiti vs. Chile, Iran vs. Japan, Nepal vs. New Zealand, Turkey vs. Italy). Each pair is compared via **stat cards**, a radar chart of normalized indicators, and an earthquake timeline. This section gives agency to the reader after the guided narrative.



CONSIDERATIONS

Here we talk about some of the charts we would have liked to further implement:

- **Recovery timeline chart** – our project measures deaths but not recovery speed. A chart showing GDP vs. time-to-rebuild (using post-earthquake GDP dip and rebound data) would extend the argument from "wealth saves lives" to "wealth also determines how fast you recover."
- **Aftershock vulnerability map** – our data covers primary events, but low-income countries are disproportionately affected by aftershocks too, because damaged buildings aren't condemned or repaired quickly. A small multiples map of aftershock death ratios would reinforce the same thesis from a different angle.

PEER ASSESSMENT

- Tamar Alpaidze
Data pipeline, EDA notebook, correlation analysis, deployment, process book.
- Manuel Curnis
Globe visualization, Gapminder decades chart, scroll manager, responsive design, parallel coordinates chart.
- Alessandro Di Maria
Hook section, dot strip (factors), country comparison with radar chart, visual identity & CSS, label de-collision.